Notebook 2 — Chapter 2.3

Order Statistics & Non-Parametric Extremes

20 worked examples — distributions of extremes & uniform spacings — calibrated for Stage 1 speed and Stage 2 trading/auction intuition

A. Distributions of Extremes

Example 1 — CDF of the Sample Maximum

$n = 5$ i.i.d. Unif$(0,1)$ draws. Find $\mathbb{P}(\max \le 0.9)$.

Solution. $\mathbb{P}(\max \le x) = [F(x)]^n$. Here $[0.9]^5 = \boxed{0.59049}$.

Example 2 — PDF of the Maximum via Differentiation

Derive the density of $\max(X_1, \dots, X_n)$ for $X_i \overset{\text{iid}}{\sim} \text{Unif}(0,1)$.

Solution. Differentiate $[F(x)]^n$: $f_{\max}(x) = n[F(x)]^{n-1} f(x) = n\,x^{n-1}$, $x \in [0,1]$.

⋆ *Intuition:* this is exactly Beta$(n,1)$ — the general rule $X_{(n)} \sim \text{Beta}(n,1)$ for uniform samples is worth memorizing outright.

Example 3 — CDF/PDF of the Sample Minimum

General formula: $\mathbb{P}(\min \ge x) = [1 - F(x)]^n \Rightarrow f_{\min}(x) = n[1 - F(x)]^{n-1} f(x)$. State the Unif$(0,1)$ case.

Solution. $f_{\min}(x) = n(1-x)^{n-1}$, $x \in [0,1]$ — this is $\boxed{\text{Beta}(1,n)}$, the mirror image of Example 2.

Example 4 — Minimum of Exponentials, Confirmed via the General Formula

$X_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$. Use the order-statistic formula to find the distribution of $\min(X_1, \dots, X_n)$.

Solution. $f_{\min}(x) = n[e^{-\lambda x}]^{n-1} \lambda e^{-\lambda x} = n\lambda e^{-n\lambda x} \Rightarrow \min \sim \boxed{\text{Exp}(n\lambda)}$.

⋆ *Intuition:* this reconfirms the "racing exponentials" result from Chapter 1.3 — order-statistic theory and competing-clocks intuition are the same fact viewed two ways.

Example 5 — Joint Density of Min and Max, and a Speed-Trick Shortcut

$n = 3$ i.i.d. Unif$(0,1)$. Find $\mathbb{P}(\min > 0.2,\ \max < 0.8)$.

Solution (general formula). $f_{(1),(n)}(a,b) = n(n-1)[F(b) - F(a)]^{n-2} f(a) f(b)$, $a < b$. For $n = 3$: $f(a,b) = 6(b-a)$. Integrating over $0.2 < a < b < 0.8$ gives 0.216.

▶ *Speed trick:* the event is identical to "all 3 values land in $(0.2, 0.8)$," an interval of length 0.6 — so $\mathbb{P} = (0.6)^3 = \boxed{0.216}$ directly, no joint density required. Reserve the joint-density formula for when min and max must satisfy *different*, asymmetric constraints.

Example 6 — The Range $R = \max - \min$

For n i.i.d. Unif$(0,1)$, find the density of R and $\mathbb{E}[R]$.

Solution. $f_R(r) = n(n-1) r^{n-2}(1-r)$, $0 \le r \le 1$ (integrate the joint density over all (a,b) with $b - a = r$). This is Beta$(n-1, 2)$, so

$$\mathbb{E}[R] = \frac{n-1}{n-1+2} = \boxed{\frac{n-1}{n+1}}.$$

⋆ *Intuition:* as $n \to \infty$, $\mathbb{E}[R] \to 1$ — with enough draws, the sample essentially spans the whole interval.

Example 7 — $\mathbb{E}[\max]$ via the Survival-Function Shortcut

Re-derive $\mathbb{E}[\max]$ for n i.i.d. Unif$(0,1)$ *without* differentiating first.

Solution. For non-negative bounded X: $\mathbb{E}[X] = \int_0^1 \mathbb{P}(X > x)\,dx$.

$$\mathbb{E}[\max] = \int_0^1 (1 - x^n)\,dx = 1 - \frac{1}{n+1} = \boxed{\frac{n}{n+1}}.$$

▶ *Speed trick:* integrating 1 − CDF is almost always faster than differentiating to get the PDF and then integrating $x\,f(x)$ — use it whenever you only need the mean.

Example 8 — The General k-th Order Statistic Density

Find the density of the median of $n = 5$ i.i.d. Unif$(0,1)$ draws ($k = 3$).

Solution. General formula: $f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!}[F(x)]^{k-1}[1 - F(x)]^{n-k} f(x)$.

$$f_{(3)}(x) = \frac{5!}{2!\,2!} x^2 (1-x)^2 = 30\,x^2(1-x)^2 = \boxed{\text{Beta}(3,3)}$$

By symmetry of Beta$(3,3)$, median $= \mathbb{E}[X_{(3)}] = 0.5$, matching $k/(n+1) = 3/6$.

Example 9 — First-Price vs. Second-Price (Vickrey) Auctions

$n = 5$ bidders, valuations i.i.d. Unif$(0,1)$. Find the expected winning valuation, and the expected price paid under truthful second-price bidding.

Solution. Winning valuation $= \max = X_{(5)}$: $\mathbb{E} = 5/6 \approx \boxed{0.833}$. Second-price payment $= X_{(4)}$ (second-highest), using $k/(n+1)$ with $k = 4$: $\mathbb{E} = 4/6 \approx \boxed{0.667}$.

⋆ *Intuition:* in a Vickrey auction, truthful bidding is a dominant strategy, so the winner's *value* is the max order statistic, but the *price* they pay is only the second-highest — the gap (here $0.833 - 0.667 = 0.167$) is the winner's expected surplus.

Example 10 — Worst-Case Fill: How Close to the Floor Is the Minimum?

$n = 20$ independent (normalized) execution slippages, Unif$(0,1)$. Find c such that $\mathbb{P}(\min > c) \ge 0.95$.

Solution. $(1-c)^{20} \ge 0.95 \Rightarrow 1 - c \ge 0.95^{1/20} \approx 0.99744 \Rightarrow c \le \boxed{0.00256}$.

⋆ *Intuition:* even with only 20 draws, you can be 95% confident the worst of them lies within 0.0026 of the absolute floor — extremes concentrate near the boundary far faster than intuition suggests. A risk desk sizing "worst-of-n" guarantees should expect the bound to be *very* tight to the edge.

B. Uniform Spacings

Example 11 — Spacings Sum to 1, and Are Identically Distributed

n points break $[0,1]$ into $n+1$ spacings $S_1, \dots, S_{n+1}$. Argue that every S_i has the same marginal distribution.

Solution. $\sum_{i=1}^{n+1} S_i = 1$ trivially. Picture the n points instead on a *circle* of circumference 1 (equivalent construction) — rotational symmetry makes all $n+1$ arcs exchangeable, so each S_i is identically (though not independently) distributed.

Example 12 — $\mathbb{E}[X_{(k)}] = k/(n+1)$, the Elegant Way

Derive $\mathbb{E}[X_{(k)}]$ using the spacings, with no integration.

Solution. $X_{(k)} = S_1 + S_2 + \cdots + S_k$. By Example 11, $\mathbb{E}[S_i] = \frac{1}{n+1}$ for every i (since $(n+1)\mathbb{E}[S_i] = \mathbb{E}[\sum S_i] = 1$). By linearity:

$$\mathbb{E}[X_{(k)}] = k \cdot \mathbb{E}[S_i] = \boxed{\frac{k}{n+1}}.$$

▶ *Speed trick:* this is dramatically faster than integrating $x \cdot f_{(k)}(x)$ from Example 8's Beta density — the spacing-sum argument is the single highest-leverage trick in this chapter.

Example 13 — Confirming via the Beta Distribution

Verify Example 12 using $X_{(k)} \sim \text{Beta}(k, n-k+1)$.

Solution. Beta(a,b) has mean $\frac{a}{a+b}$. Here $a = k$, $b = n - k + 1$:

$$\mathbb{E}[X_{(k)}] = \frac{k}{k + (n-k+1)} = \boxed{\frac{k}{n+1}}. \ \checkmark$$

Example 14 — Variance of the k-th Order Statistic

Find $\text{Var}(X_{(k)})$ for $n = 9$, $k = 5$ (the median of 9 uniforms).

Solution. Beta$(k, n-k+1)$ variance: $\text{Var} = \frac{k(n-k+1)}{(n+1)^2(n+2)}$.

$$\text{Var}(X_{(5)}) = \frac{5 \cdot 5}{(10)^2(11)} = \frac{25}{1100} \approx \boxed{0.0227}, \quad \text{sd} \approx 0.151.$$

Example 15 — Rényi's Exponential Representation

State the representation connecting uniform spacings to i.i.d. Exponentials, and note why it matters.
Solution. Let $E_1, \dots, E_{n+1} \overset{\text{iid}}{\sim} \text{Exp}(1)$. Then

$$(S_1, \dots, S_{n+1}) \overset{d}{=} \left(\frac{E_1}{\sum_j E_j}, \dots, \frac{E_{n+1}}{\sum_j E_j} \right).$$

⋆ *Intuition:* normalized i.i.d. Exponentials are exchangeable and sum to 1 by construction — exactly matching Example 11's requirements. This representation turns hard "broken stick" probability questions into Gamma/Exponential calculations (used in Examples 16–17 and the capstone, Example 20).

Example 16 — All Spacings Small: Rederiving the Triangle Probability

$n = 2$ break points (3 pieces). Using the general "all spacings $\le c$" formula

$$\mathbb{P}\left(\max_i S_i \le c\right) = \sum_{j=0}^{\lfloor 1/c \rfloor} (-1)^j \binom{n+1}{j} (1 - jc)_+^n,$$

compute $\mathbb{P}(\max_i S_i \le \frac{1}{2})$.
Solution. $n = 2$, $c = \frac{1}{2}$, $\lfloor 1/c \rfloor = 2$:

$$1 - \binom{3}{1}(0.5)^2 + \binom{3}{2}(0)^2 = 1 - 0.75 + 0 = \boxed{0.25}.$$

⋆ *Intuition:* "all 3 pieces $\le \frac{1}{2}$" is *exactly* the triangle inequality for a stick broken at 2 random points — this independently reconfirms Chapter 1.3's $\mathbb{P}(\text{triangle}) = 1/4$ by a completely different method.

Example 17 — The Same Formula, Beyond the Triangle Case

$n = 4$ break points (5 pieces). Find $\mathbb{P}(\max_i S_i \le 0.3)$.
Solution. $\lfloor 1/0.3 \rfloor = 3$:

$$\binom{5}{0}(0.7)^0 \cdot 1 - \binom{5}{1}(0.7)^4 + \binom{5}{2}(0.4)^4 - \binom{5}{3}(0.1)^4$$

$$= 1 - 5(0.2401) + 10(0.0256) - 10(0.0001)$$

$$= 1 - 1.2005 + 0.256 - 0.001 = \boxed{0.0545}.$$

Only $\approx 5.5\%$ chance all 5 pieces stay under 0.3 — illustrating how quickly "every piece small" becomes unlikely as pieces outnumber the threshold's capacity ($5 \times 0.3 = 1.5 > 1$, so it's *possible* but tight).

Example 18 — Covariance Between Spacings, via the "Sum Is Constant" Trick

Find $\text{Cov}(S_i, S_j)$ for $i \neq j$.
Solution. Since $\sum_{i=1}^{n+1} S_i = 1$ (a constant), $\text{Var}(\sum S_i) = 0$:

$$0 = (n+1)\,\text{Var}(S_i) + (n+1)n\,\text{Cov}(S_i, S_j)$$

S_i itself is Beta$(1, n)$ (Example 3), so $\text{Var}(S_i) = \dfrac{n}{(n+1)^2(n+2)}$. Solving:

$$\text{Cov}(S_i, S_j) = -\frac{\text{Var}(S_i)}{n} = \boxed{-\frac{1}{(n+1)^2(n+2)}}.$$

▶ *Speed trick:* whenever a group of variables sums to a constant, Var(sum) = 0 instantly gives you every pairwise covariance for free — no joint density needed.

Example 19 — Trading Analogy: Expected Gap Between Order Arrivals

$m = 9$ independent order-arrival times, uniform over a $T = 390$-minute trading session. Find the expected gap between consecutive arrivals (including the open/close buffers).
Solution. 9 points create 10 spacings on $[0, T]$; by Example 11 (rescaled), each has mean $T/(m+1)$:

$$\mathbb{E}[\text{gap}] = \frac{390}{10} = \boxed{39 \text{ minutes}}.$$

Example 20 — Capstone: Expected Value of the Largest Spacing

Find $\mathbb{E}[\max_i S_i]$ exactly, for general n.
Solution. Using Example 15's representation, $S_i = E_i/T$ where $T = \sum_j E_j \sim \text{Gamma}(n+1, 1)$, and T is **independent** of the ratio vector $(S_1, \dots, S_{n+1})$ (a classical Gamma–Dirichlet fact). Since $\max_i S_i$ depends only on the ratios, it is independent of T, and $\max_i E_i = T \cdot \max_i S_i$, so

$$\mathbb{E}[\max_i E_i] = \mathbb{E}[T] \cdot \mathbb{E}[\max_i S_i] \;\Rightarrow\; \mathbb{E}[\max_i S_i] = \frac{\mathbb{E}[\max_i E_i]}{\mathbb{E}[T]} = \boxed{\frac{H_{n+1}}{n+1}}$$

using $\mathbb{E}[\max \text{ of } n{+}1 \text{ iid Exp}(1)] = H_{n+1}$ (Chapter 1.3/2.2) and $\mathbb{E}[T] = n+1$.
⋆ *Intuition — full circle:* at $n = 1$ this gives $H_2/2 = 0.75$, exactly matching Chapter 1.3's $\max(X, 1-X) \sim \text{Unif}(0.5, 1)$ result. Three chapters' worth of machinery (racing Exponentials, harmonic numbers, uniform spacings) collapse into one clean closed form.

Drill order: re-derive Examples 5, 12, 16, 18, and 20 from scratch, closed-book — they are the five "speed trick" results (shortcut over brute force, sum-is-constant, and the Rényi capstone) that carry this entire chapter's exam value.